

20 March 2023 18:06

bell curve

→ pdf



- ✓

Denoted as:

$\mu \rightarrow \text{mean}$
 $\sigma \rightarrow \text{std}$

No small



data $\rightarrow$ Normal $\rightarrow \boxed{\mu, \sigma}$

inter $\rightarrow$ $\sigma \sqrt{2\pi}$

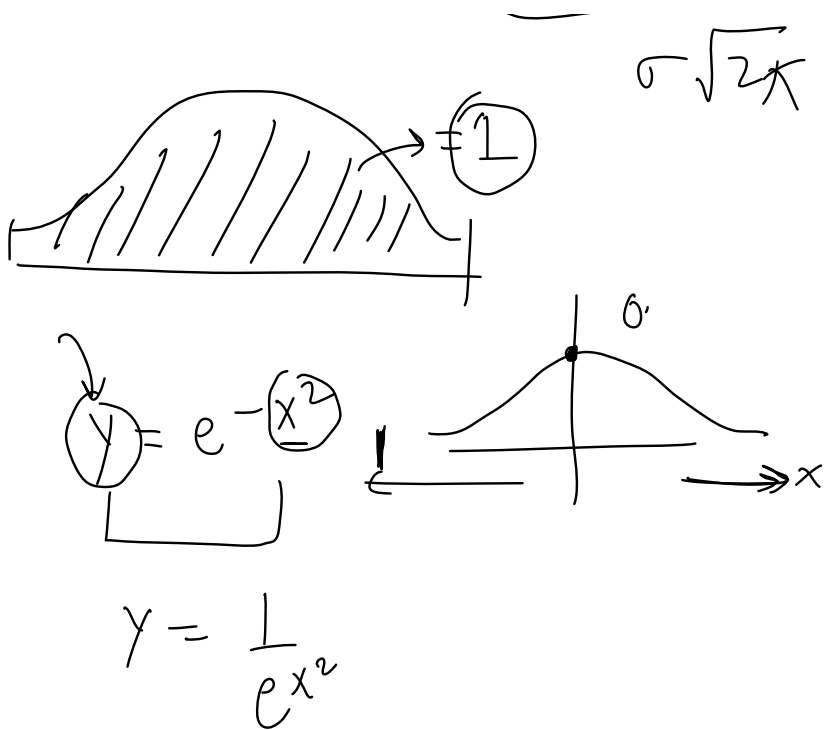
The diagram illustrates the relationship between data and inter. It shows two overlapping normal distribution curves. An arrow points from 'data' to the curves, and another arrow points from the curves to a box containing μ and σ . A third arrow points from 'inter' to the width of the curves, which is labeled $\sigma \sqrt{2\pi}$.

4

Equation in detail:

Equation in detail:

$$Y = e^{-\frac{(x-\mu)^2}{2\sigma^2}} = \frac{e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}}{\sigma\sqrt{2\pi}}$$



Standard Normal Variate (Z) → Standard Normal distribution

20 March 2023 18:08

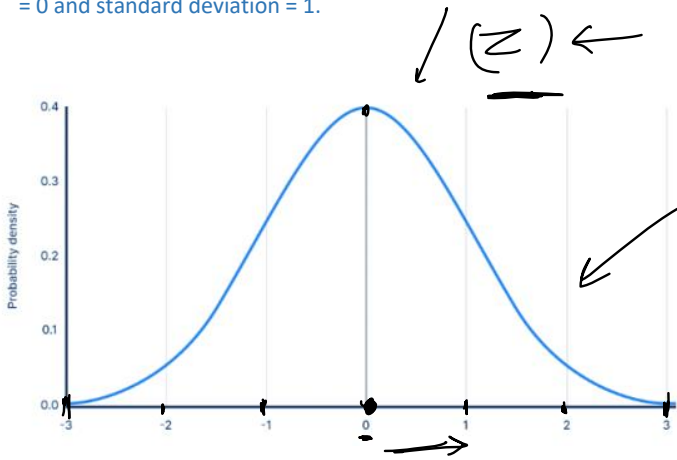
$$X \sim N(\mu, \sigma) \quad \mu=0 \quad \sigma=1$$

$$\downarrow$$

$$Z \sim N(0, 1)$$

What is Standard Normal Variate

A Standard Normal Variate (Z) is a standardized form of the normal distribution with mean = 0 and standard deviation = 1.



Standardizing a normal distribution allows us to compare different distributions with each other, and to calculate probabilities using standardized tables or software.

Equation:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

$X \sim N(5, 2.5)$

How to transform a normal distribution to Standard Normal Variate

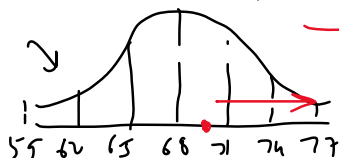
$$X \sim N(29, 14)$$

Refer Python code

Kya Fayda Standardize karne ka?

Suppose the heights of adult males in a certain population follow a normal distribution with a mean of 68 inches and a standard deviation of 3 inches. What is the probability that a randomly selected adult male from this population is taller than 72 inches?

$$X \sim N(68, 3)$$



What are Z-tables

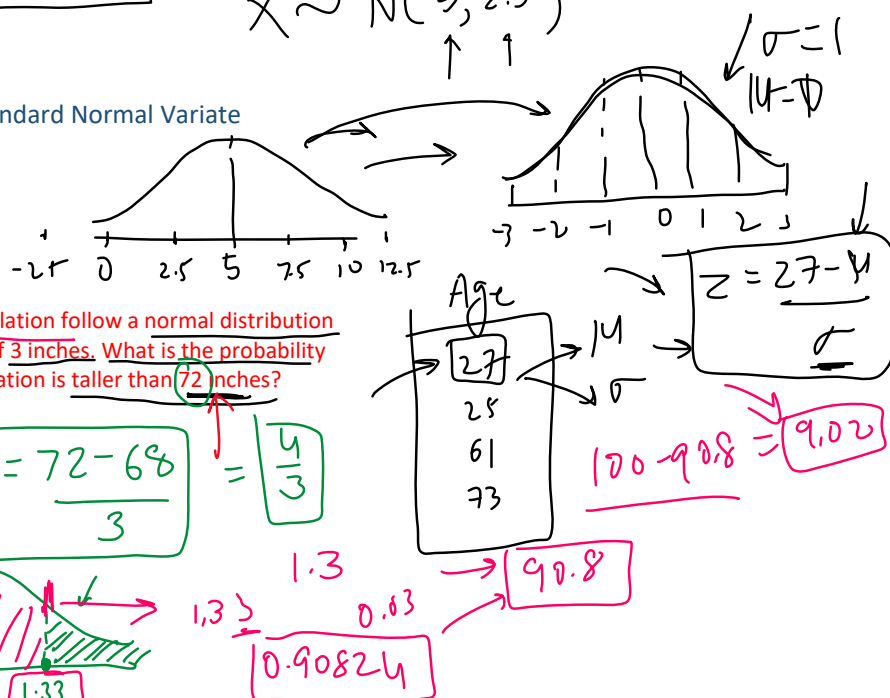
A z-table tells you the area underneath a normal distribution curve, to the left of the z-score

<https://www.ztable.net/>

For a Normal Distribution $X \sim (\mu, \text{std})$ what percent of population lie between mean and 1 standard deviation, 2 std and 3 std?

$$X \sim N(\mu, \sigma) \quad \mu \rightarrow \quad \sigma \rightarrow$$

$$Z = \frac{X - \mu}{\sigma}$$



$$X \sim N(\mu, \sigma)$$

$$Z = \frac{X - \mu}{\sigma}$$

$$Z = \frac{\mu - \mu}{\sigma} = 0$$

$$Z = \frac{X - \mu}{\sigma} = \frac{\mu + \sigma - \mu}{\sigma}$$

$$0.9772$$

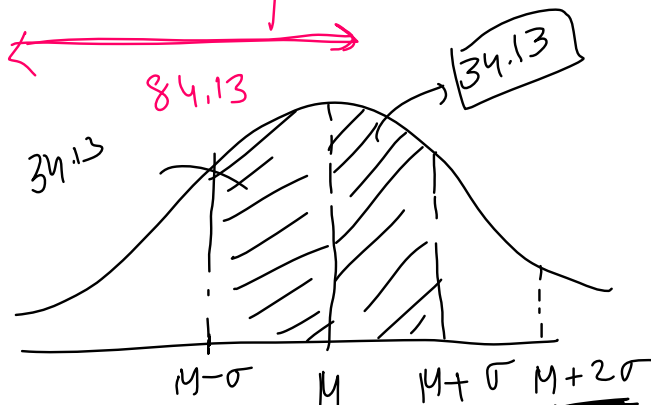
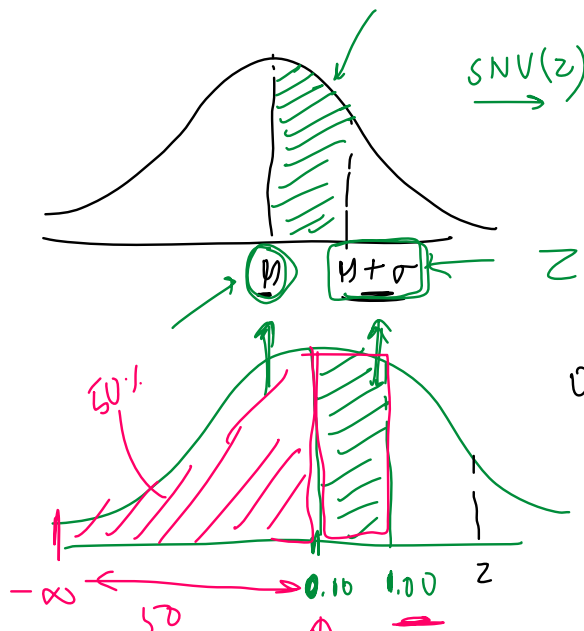
$$97.72\%$$

$$= \frac{\sigma}{\sigma} = 1$$

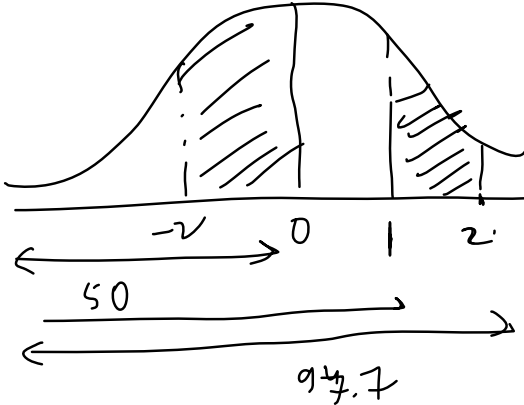
$$84.13 - 50 = 34.13$$

$$-1\sigma \quad +1\sigma$$

$$64.28\%$$



$$Z = \frac{\mu + 2\sigma - \mu}{\sigma} = 2$$

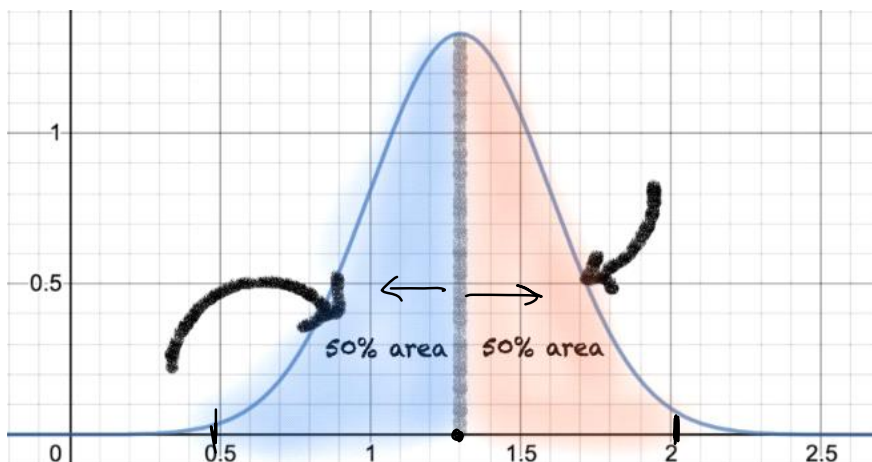


Properties of Normal Distribution

20 March 2023 18:06

1. Symmetry

The normal distribution is symmetric about its mean which means that the probability of observing a value above the mean is the same as the probability of observing a value below the mean. The bell-shaped curve of the normal distribution reflects this symmetry.



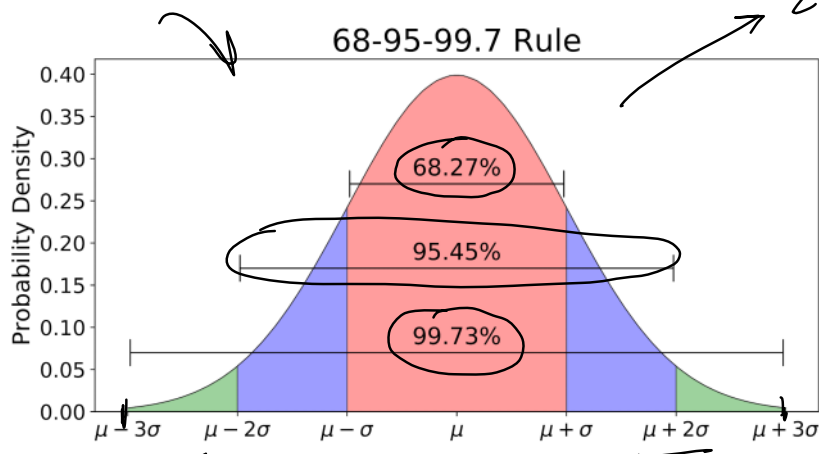
2. Measures of Central Tendencies are equal $\rightarrow$ mean $\rightarrow$ median $\rightarrow$ mode

3. Empirical Rule

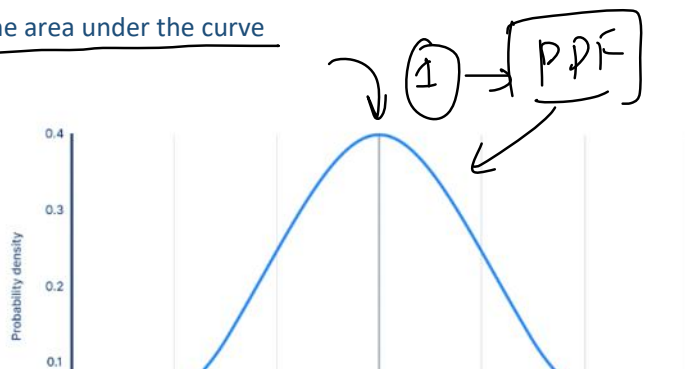
The normal distribution has a well-known empirical rule, also called the 68-95-99.7 rule, which states that approximately 68% of the data falls within one standard deviation of the mean, about 95% of the data falls within two standard deviations of the mean, and about 99.7% of the data falls within three standard deviations of the mean.

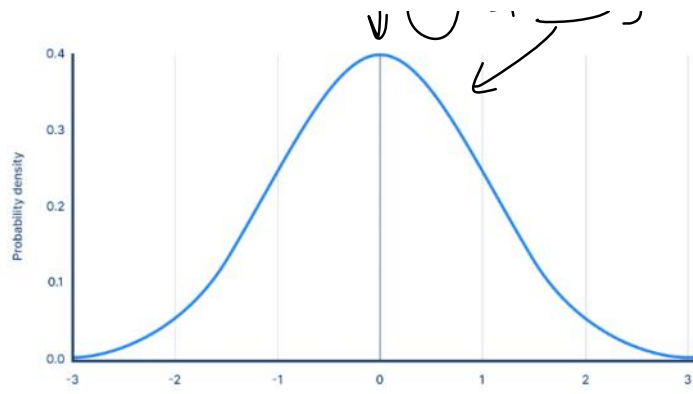
Standard deviation

Z-table



4. The area under the curve



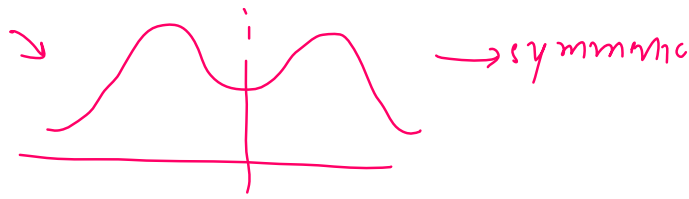


Skewness

20 March 2023 18:07

What is skewness?

A normal distribution is a bell-shaped, symmetrical distribution with a specific mathematical formula that describes how the data is spread out. Skewness indicates that the data is not symmetrical, which means it is not normally distributed.

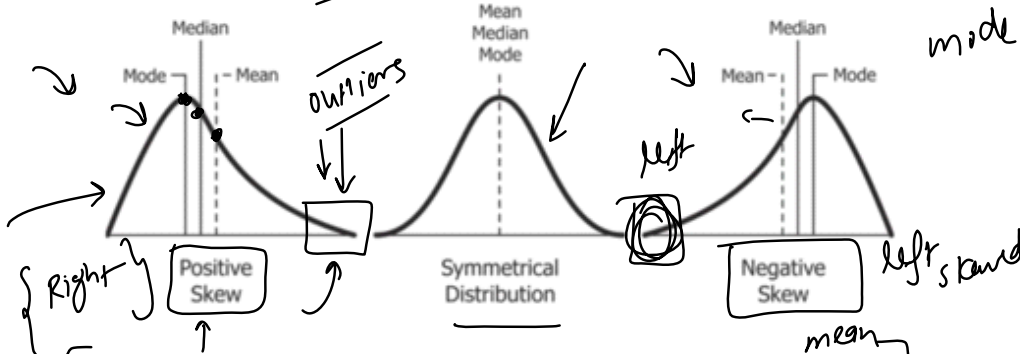


Skewness is a measure of the asymmetry of a probability distribution. It is a statistical measure that describes the degree to which a dataset deviates from the normal distribution.

In a symmetrical distribution, the mean, median, and mode are all equal. In contrast, in a skewed distribution, the mean, median, and mode are not equal, and the distribution tends to have a longer tail on one side than the other.

Skewness can be positive, negative, or zero. A positive skewness means that the tail of the distribution is longer on the right side, while a negative skewness means that the tail is longer on the left side. A zero skewness indicates a perfectly symmetrical distribution.

mode < median < mean
tail event



moment
mode > median > mean
2 moment - variance
3 moment - skewness
4th - kurtosis

The greater the skew the greater the distance between mode, median and mean

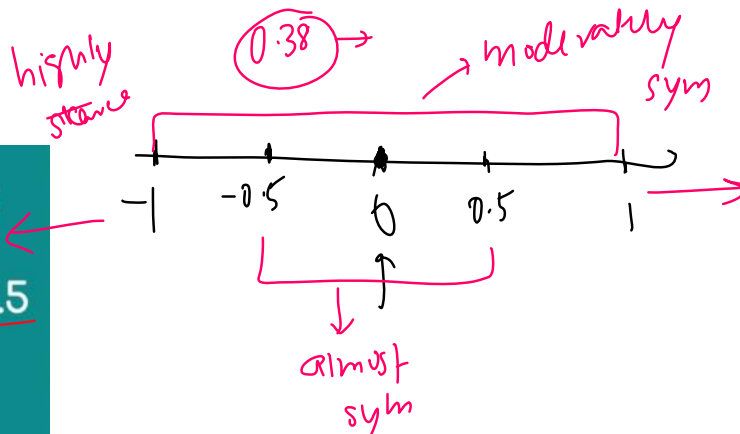
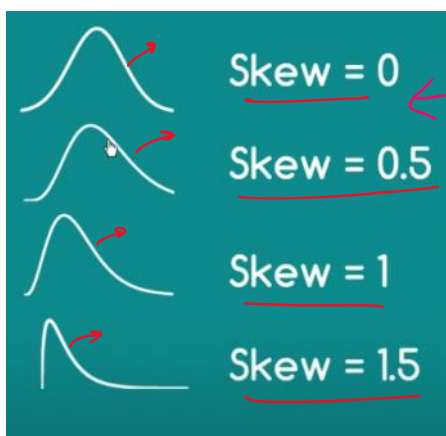
How skewness is calculated?

$$\frac{n}{(n-1)(n-2)} \sum \left(\frac{(x - \bar{x})}{s} \right)^3$$

moment
sample skew

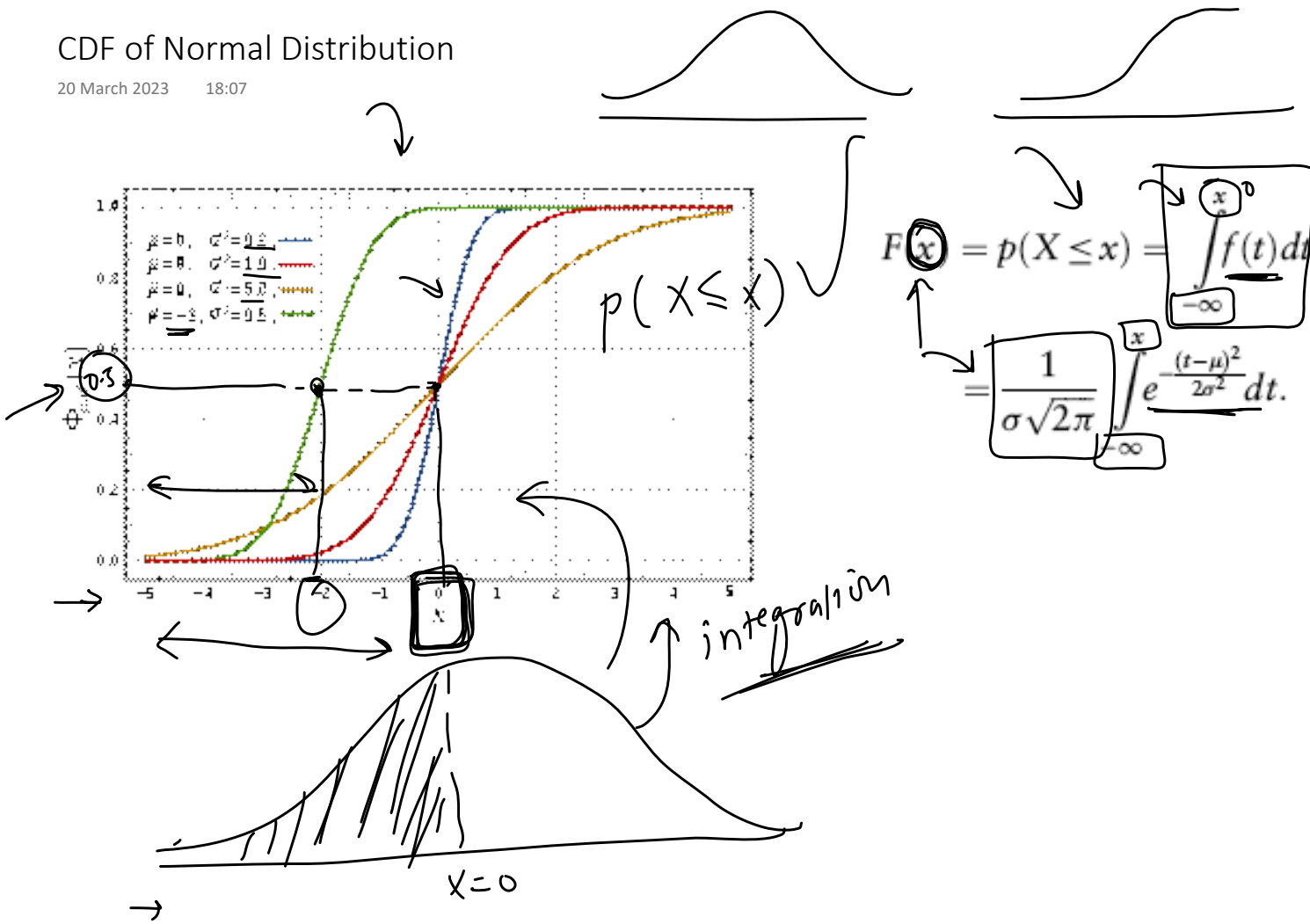
Python Example

Interpretation



CDF of Normal Distribution

20 March 2023 18:07



Use in Data Science

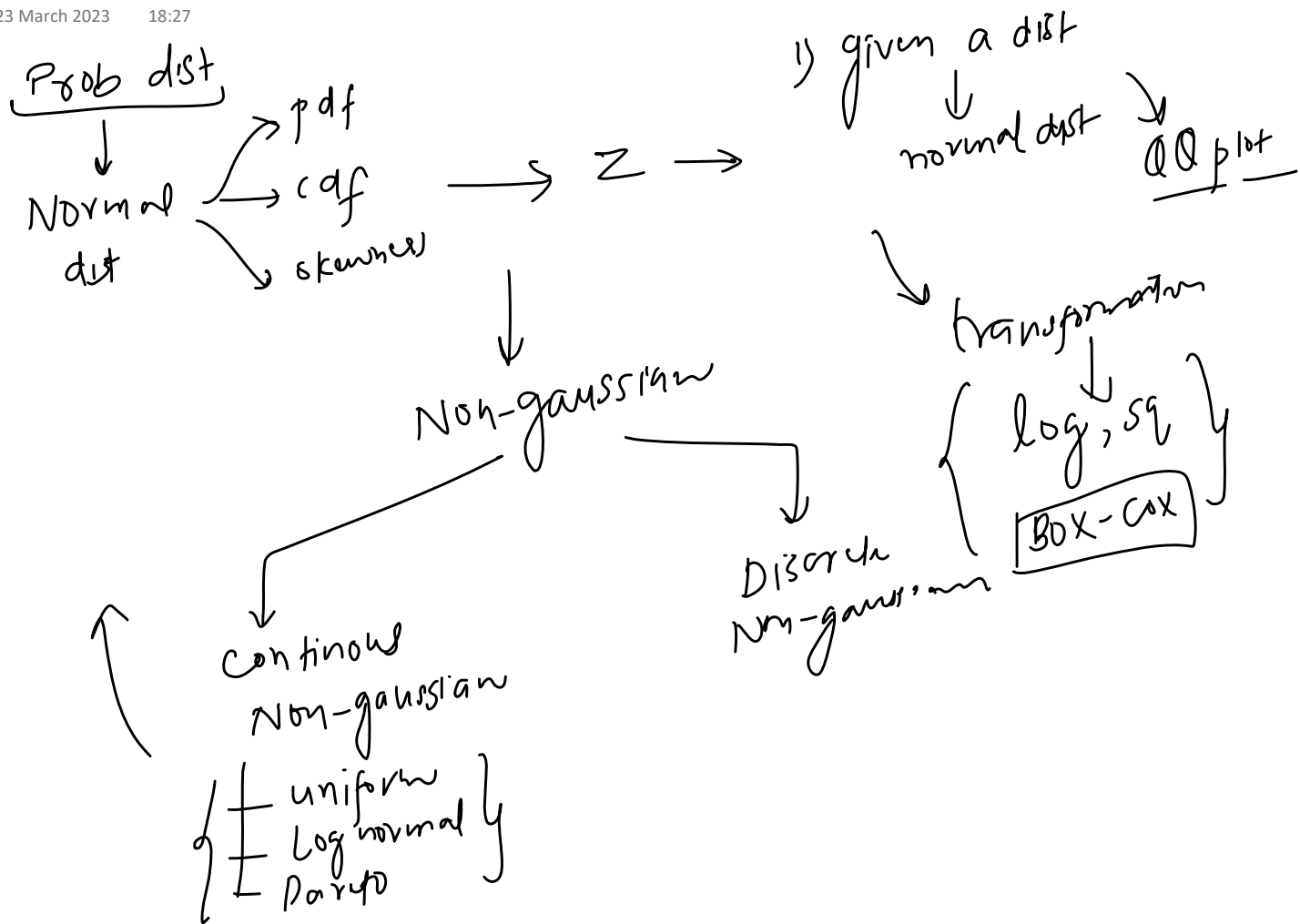
20 March 2023

18:08

- 
- 
- Outlier detection
 - Assumptions on data for ML algorithms -> Linear Regression and GMM
 - Hypothesis Testing
 - Central Limit Theorem

Recap

23 March 2023 18:27



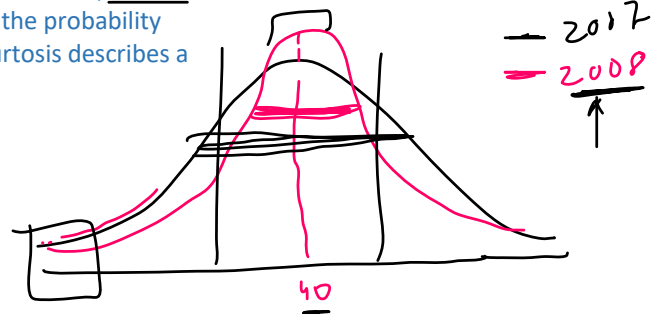
Kurtosis

23 March 2023 13:18

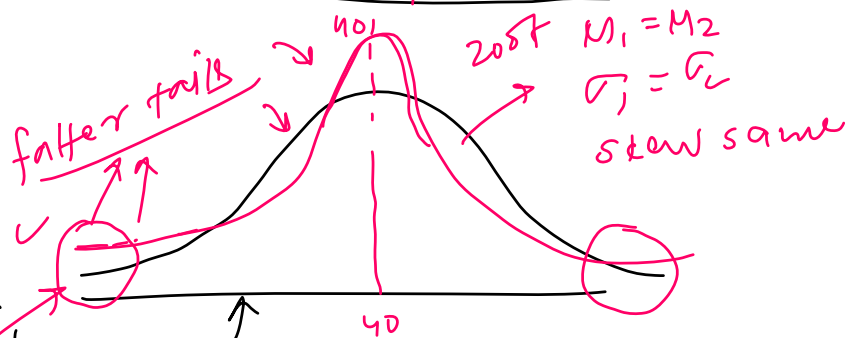
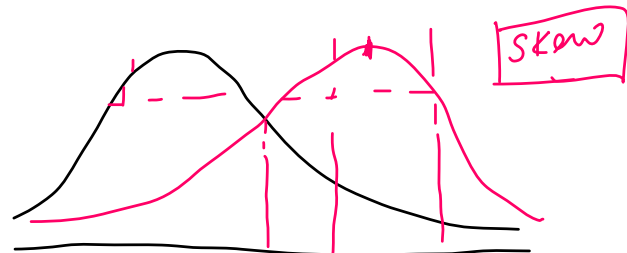
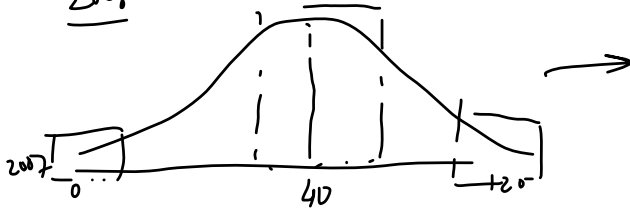
What is Kurtosis?

Kurtosis is the 4th statistical moment. In probability theory and statistics, kurtosis (meaning "curved, arching") is a measure of the "tailedness" of the probability distribution of a real-valued random variable. Like skewness, kurtosis describes a particular aspect of a probability distribution.

1st → mean
2nd → std
3rd → skewness
4th → kurtosis



Batman → Sachin
2007 → 100 matches → 40 avg
2008 → 100 matches → 40 avg



False notation about Kurtosis

<https://en.wikipedia.org/wiki/Kurtosis>

Formula

sample kurtosis

$$\left\{ \frac{n * (n+1)}{(n-1) * (n-2) * (n-3)} * \sum_i^n \left(\frac{x_i - \bar{x}}{s} \right)^4 \right\} - \frac{3 * (n-1)^2}{(n-2) * (n-3)}$$

Practical Use-case

In finance, kurtosis risk refers to the risk associated with the possibility of extreme outcomes or "fat tails" in the distribution of returns of a particular asset or portfolio.

If a distribution has high kurtosis, it means that there is a higher likelihood of extreme events occurring, either positive or negative, compared to a normal distribution.

In finance, kurtosis risk is important to consider because it indicates that there is a greater probability of large losses or gains occurring, which can have significant implications for investors. As a result, investors may want to adjust their investment strategies to account for kurtosis risk.



Excess Kurtosis & Types

Excess kurtosis is a measure of how much more peaked or flat a distribution is

Excess Kurtosis & Types

Excess kurtosis is a measure of how much more peaked or flat a distribution is compared to a normal distribution, which is considered to have a kurtosis of 0. It is calculated by subtracting 3 from the sample kurtosis coefficient.

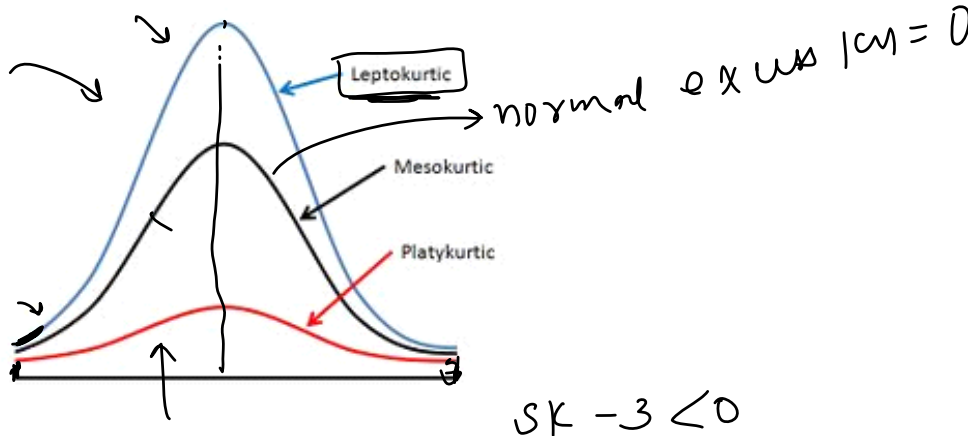
Types of Kurtosis

$$\text{sample } k - 3 > 0$$

Leptokurtic

A distribution with positive excess kurtosis is called leptokurtic. "Lepto-" means "slender". In terms of shape, a leptokurtic distribution has fatter tails. This indicates that there are more extreme values or outliers in the distribution.

Example - Assets with positive excess kurtosis are riskier and more volatile than those with a normal distribution, and they may experience sudden price movements that can result in significant gains or losses.



Platykurtic

A distribution with negative excess kurtosis is called platykurtic. "Platy-" means "broad". In terms of shape, a platykurtic distribution has thinner tails. This indicates that there are fewer extreme values or outliers in the distribution.

Assets with negative excess kurtosis are less risky and less volatile than those with a normal distribution, and they may experience more gradual price movements that are less likely to result in large gains or losses.

Mesokurtic

Distributions with zero excess kurtosis are called mesokurtic. The most prominent example of a mesokurtic distribution is the normal distribution family, regardless of the values of its parameters.

Mesokurtic is a term used to describe a distribution with a excess kurtosis of 0, indicating that it has the same degree of "peakedness" or "flatness" as a normal distribution.

Example -

In finance, a mesokurtic distribution is considered to be the ideal distribution for assets or portfolios, as it represents a balance between risk and return.

QQ Plot

23 March 2023 13:19

- How to find if a given distribution is normal or not?

Visual inspection One of the easiest ways to check for normality is to visually inspect a histogram or a density plot of the data. A normal distribution has a bell-shaped curve, which means that the majority of the data falls in the middle, and the tails taper off symmetrically. If the distribution looks approximately bell-shaped, it is likely to be normal.

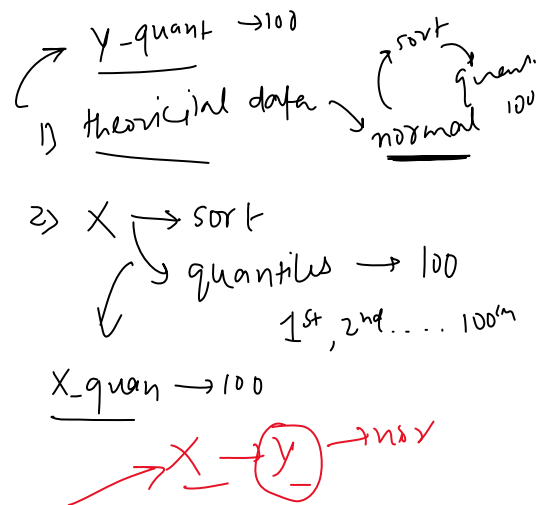
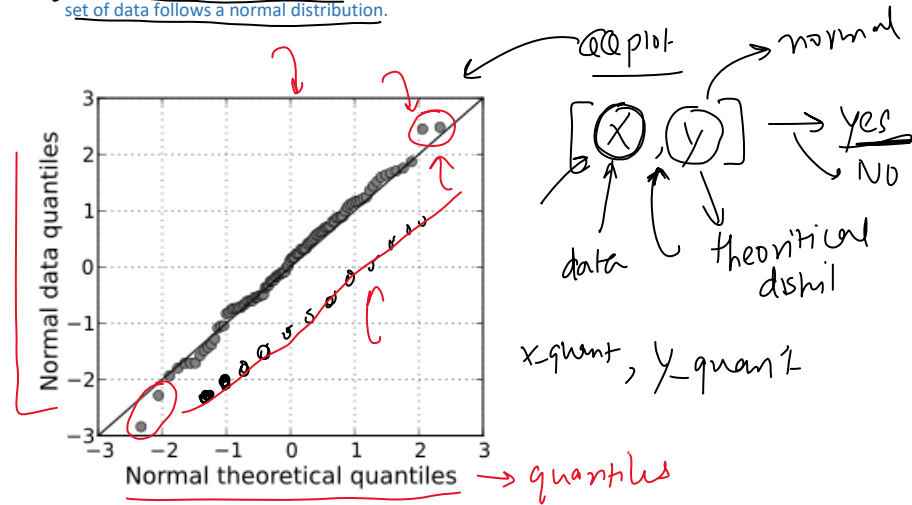
QQ Plot Another way to check for normality is to create a normal probability plot (also known as a Q-Q plot) of the data. A normal probability plot plots the observed data against the expected values of a normal distribution. If the data points fall along a straight line, the distribution is likely to be normal.

Statistical tests There are several statistical tests that can be used to test for normality, such as the Shapiro-Wilk test, the Anderson-Darling test, and the Kolmogorov-Smirnov test. These tests compare the observed data to the expected values of a normal distribution and provide a p-value that indicates whether the data is likely to be normal or not. A p-value less than the significance level (usually 0.05) suggests that the data is not normal.



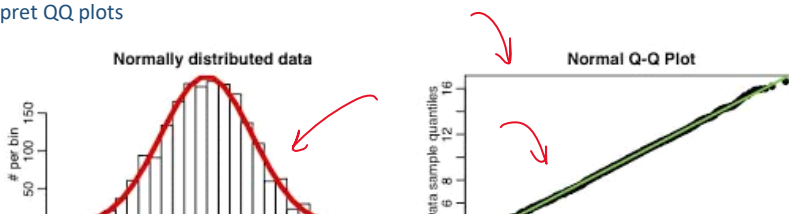
- What is a QQ Plot and how is it plotted?

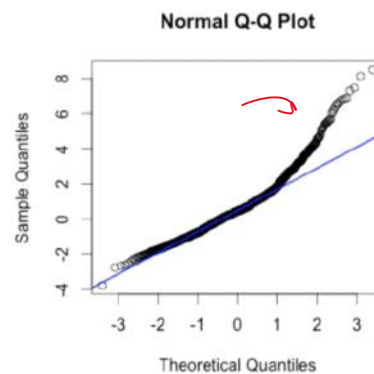
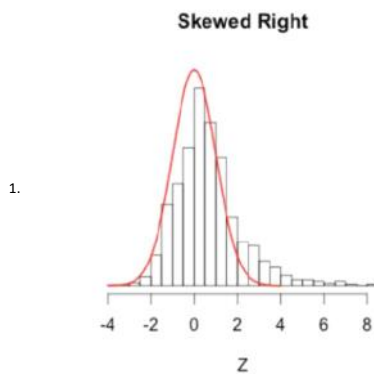
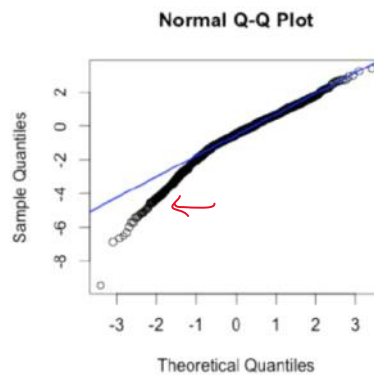
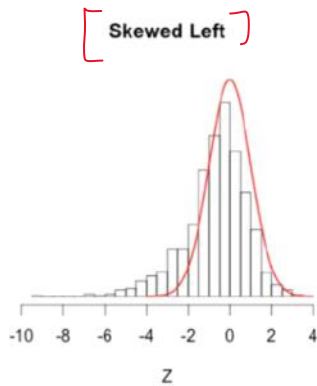
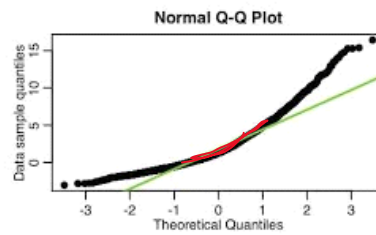
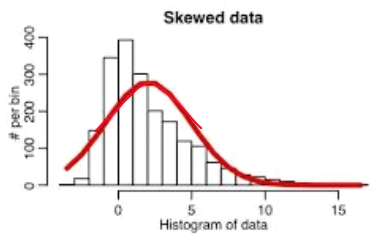
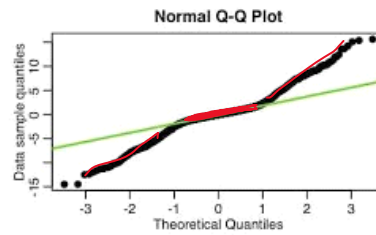
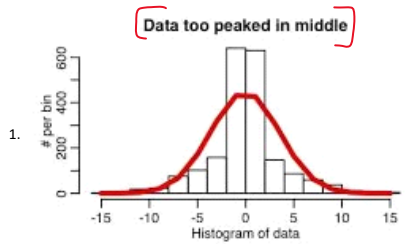
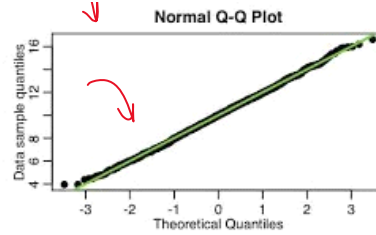
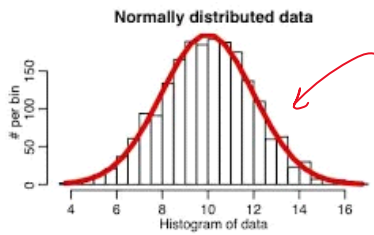
A QQ plot (quantile-quantile plot) is a graphical tool used to assess the similarity of the distribution of two sets of data. It is particularly useful for determining whether a set of data follows a normal distribution.

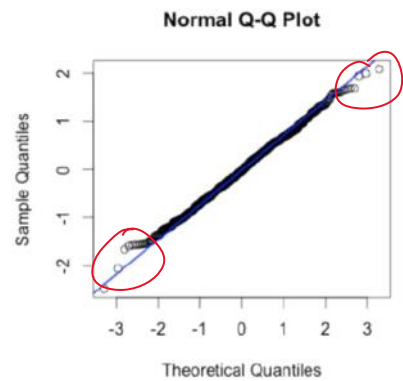
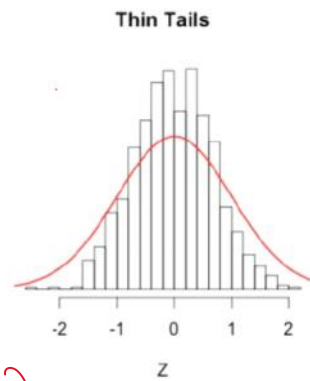
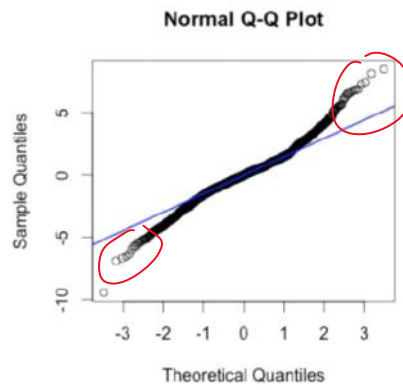
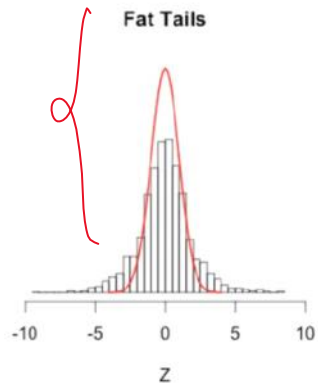


In a QQ plot, the quantiles of the two sets of data are plotted against each other. The quantiles of one set of data are plotted on the x-axis, while the quantiles of the other set of data are plotted on the y-axis. If the two sets of data have the same distribution, the points on the QQ plot will fall on a straight line. If the two sets of data do not have the same distribution, the points will deviate from the straight line.

- Python example ✓
- <https://www.statsmodels.org/dev/generated/statsmodels.graphics.gofplots.qqplot.html>
- How to interpret QQ plots







1.

- Does QQ plot only detect normal distribution?

